ADC Section 1

2025年11月11日 星期二 08:53

Analysis & Design of Circuits Lab. S1
24 October, 2025
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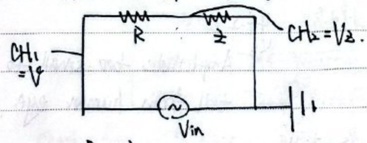
Reference. ADC section 1.

Aim: Measure impedance of resistors and parasitic impedance of signal generator.

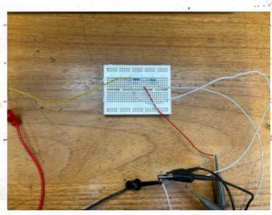
Procedure.

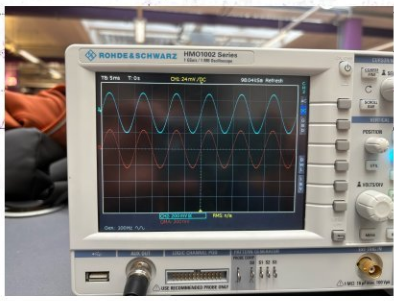
- Measuring impedance of simple resistor.
  ∵ Channels of oscilliscope can only measure V relative to ground.
  for CH1 & CH2, they are internally connected, so otherwise short circuit happens.

Measure $V_R = V - V_Z$.



when $R = Z = 1k\Omega$, $V_R = V_Z$ by using "MATH" on CRO.





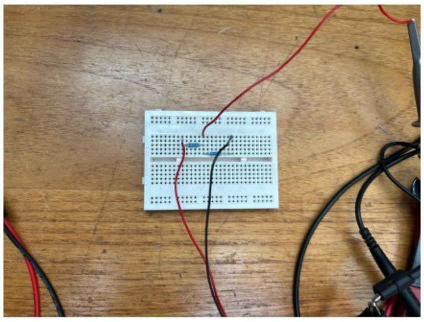
when $V_R \neq V_Z$, $Z = R \cdot \frac{V_Z}{V_R}$.

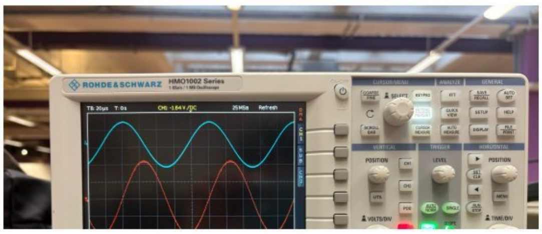
we use auto-measure and cursor to test which one is more accurate: Peak-Peak auto-measure:

V_Z/V	V_R/V	R/Ω	Z (calcus)/Ω	Z (Actuals)/Ω
2.60	2.80	1000	929	900
4.32	1.72	1000	2.51k	2.88k
3.92	2.04	1000	1.92k	2.14k
4.88	919.5m	1000	5.31k	4.68k
Cursor measure:				
2.60	3.00	1000	847	
4.27	1.55	1000	2.75k	
3.87	1.80	1000	2.15k	
4.73	700.3m	1000	6.75k.	

They give error:

Auto: Cursor measure





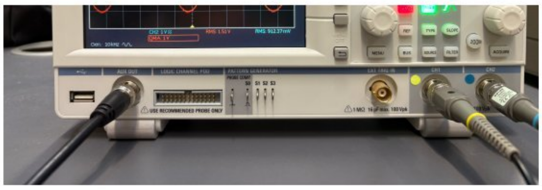
3.22%	0.333%
12.8%	4.51%
10.3%	0.467%
13.5%	44.2%

↳ Amplitude too small to tell from human eye.

So in conclusion, $Z = R\frac{V_2}{V_R}$ is true.

Auto-measure has larger error, but still reliable; it's much faster and easier.

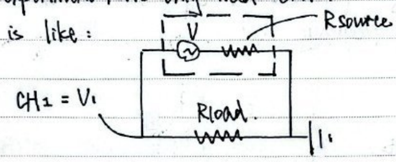
Note. In experiment I used weird numbers of resistance because I thought that was uncertainty, and I used the readings on multi-meter. Then I found when I was using multi-meter, I actually connected myself in parallel with resistor, which is the reason why resistance is not same as written (always smaller). From then on I know, and I use the written value, which proved to be accurate.



· Challenge: Measure source impedance of the signal generator.

In this experiment, we only need CH_1.

Circuit is like:



where unknown are Rsource and V.

$\because V_1 = V \cdot \frac{R_{load}}{R_{load} + R_{source}}$

V_1 / V	R_{load} / Ω
4.84	0.986k
4.08	0.198k
4.40	8.69k

$$\therefore \begin{cases} 4.08 = V \cdot \frac{198}{198 + R_{source}} \\ 4.40 = V \cdot \frac{8690}{8690 + R_{source}} \end{cases} \Rightarrow \begin{cases} R_{source} = 15.9\Omega \\ V = 4.41V \end{cases}$$

Check for 3rd pair: LHS = 4.84

$RHS = \frac{986}{15.9 + 986} \times 4.41 = 4.34$.

$\therefore$ Error $= \frac{4.84 - 4.34}{4.84} = 10.3\%$. ⇒ Medium.

which gives 15.9Ω source impedance.